

Enrique Noriega-Atala

☎ (520) 241-5945 ✉ enrique.noriega@gmail.com 💻 enoriega.info

Education

PhD	The University of Arizona , Information Science Minor: Statistics	Jan 2020
MS	The University of Arizona , Computer Science	Jan 2014
BS	Monterrey Institute of Technology and Higher Education , Information Technologies	Jan 2010

Experience

Northeastern University , Associate Research Scientist, Khoury College of Computer Sciences	Feb 2026 – present 8 months
Researcher in the Gyori Lab at Northeastern University, focusing on natural language processing and biomedical informatics with an emphasis on information extraction from scientific literature. • Develop models and datasets for information extraction in the biomedical domain. • Mentor graduate students whose focus of research includes Natural Language Processing. • Engage and liaise on behalf of Gyori lab with external research groups from the lab's research projects.	
The University of Arizona , Computational Sciences Researcher IV, Computer Science Department	Oct 2022 – Jan 2026 3 years 4 months
• Consult as NLP/LLM domain matter expert in research engagements between Computer Science and various academic departments. • Co-develop curriculum and teach workshops for graduate students, including Exploring the LLM Frontier and AI Makerspace. • Lead and collaborate on research projects about information retrieval, extraction, document modeling, program synthesis, etc. • Write and publish research work on top-tier, peer-reviewed venues. • Collaborate with planning and writing extramural research grants and other funding mechanisms. • Participate in the organization of academic workshops. • Mentor graduate students.	
The University of Arizona , Postdoctoral Research Associate, Computer Science Department	Jan 2021 – Sept 2022 1 year 9 months
• Collaborated on the development of five research publications in top-tier peer-reviewed venues. • Led a collaborative effort to develop a multi-disciplinary visual analytics tool for the visualization of information extraction in the biomedical domain. • Contributed to open source NLP projects. • Presented research projects in multiple symposiums and conferences. • Collaborated on the development of two grant proposals for NSF and DARPA. • Collaborated and coordinated the organization of two research workshops, one institutional and the other co-hosted at COLING '22. • Mentored graduate students in the Department of Computer Science.	

CodaMetrix , Senior NLP Scientist • Built a taxonomy of body regions, organs, and tissues using medical ontologies such as UMLS, SNOMED, and MeSH. • Bootstrapped a dataset using proprietary EHR corpora. • Implemented, trained, and tested state-of-the-art neural architectures with PyTorch and TensorFlow. • Developed APIs for integration with the company's product. • Participated as a mentor in the summer internship program and contributed to documentation and internal training curricula for NLP and machine learning.	Feb 2020 – Oct 2020 9 months
The University of Arizona , Graduate Research Associate Worked within CLU lab and ML4AI on NLP and Machine Learning projects applied to health sciences, contributing regularly to REACH, a machine reading system for biomedical publications.	Jan 2013 – Jan 2020 7 years 1 month
The University of Arizona , Teaching Assistant • Collaborated as a teaching assistant for graduate and undergraduate courses in Computer Science and the School of Information. • Graded assignments, exams, and projects, held office hours, designed assignments, and directed laboratory sessions and guest lectures. • Courses served: Computer Organization (2013), Data Structures (2014), Information Retrieval and Web Search (2015, 2016), Introduction to Machine Learning (2017, 2019).	Aug 2013 – Dec 2019 6 years 5 months

Honors and Awards

UA Graduate & Professional Student's Council Travel Grant	Jan 2018
UA School of Information's Travel Award	Jan 2018
International Conference in Data Mining Student Award	Jan 2018
Galileo Circle Scholar, UA's College of Science	Jan 2014
Instituto Educativo Sonora-Arizona Scholarship	Jan 2013 – Jan 2014
CONACyT Graduate Studies Scholarship	Jan 2012 – Jan 2014
CENEVAL's Outstanding Performance Testimony on the EGEL Test	Jan 2010

Journal Publications

Extracting Inter-sentence Relations for Associating Biological Context with Events in Biomedical Text <i>Enrique Noriega-Atala</i> , Paul D. Hein, Shraddha S. Thumasi, Zechy Wong, Xia Wang, Sean M. Hendryx, Clayton T. Morrison (IEEE/ACM Transactions on Computational Biology and Bioinformatics)	Jan 2019
Large-scale Automated Machine Reading Discovers New Cancer Driving Mechanisms Marco A. Valenzuela-Escárcega, Özgün Babur, Gus Hahn-Powell, Dane Bell, Thomas Hicks, <i>Enrique Noriega-Atala</i> , Xia Wang, Mihai Surdeanu, Emek Demir, Clayton T. Morrison (Database: The Journal of Biological Databases and Curation)	Jan 2018

Conference Publications

- When and Where Did it Happen? An Encoder-Decoder Model to Identify Scenario Context** Jan 2024
Enrique Noriega-Atala, Robert Vacareanu, Salena Torres Ashton, Adarsh Pyarelal, Clayton T. Morrison, Mihai Surdeanu
(Findings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP))
- Active learning design choices for NER with transformers** Jan 2024
Robert Vacareanu, *Enrique Noriega-Atala*, Gus Hahn-Powell, Marco A. Valenzuela-Escárcega, Mihai Surdeanu
(Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING))
- Visualizing Interaction Networks and Evidence in Biomedical Corpora** Jan 2023
Enrique Noriega-Atala, Md Rahat-Uz-Zaman, Ruchika Bhat, Mladen Jergović, Stephen G. Kobourov, Janko Nikolich-Žugich
(Proceedings of the 16th IEEE Pacific Visualization Symposium (PacificVis))
- A Human-machine Interface for Few-shot Rule Synthesis for Information Extraction** Jan 2022
Robert Vacareanu, George C. Barbosa, *Enrique Noriega-Atala*, Gus Hahn-Powell, Rebecca Sharp, Marco A. Valenzuela-Escárcega, Mihai Surdeanu
(Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT))
- Learning what to read: Focused machine reading** Jan 2017
Enrique Noriega-Atala, Marco A. Valenzuela-Escárcega, Clayton T. Morrison, Mihai Surdeanu
(Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP))

Workshop Publications

- Variable Extraction for Model Recovery in Scientific Literature** Jan 2025
Chunwei Liu, *Enrique Noriega-Atala*, Adarsh Pyarelal, Clayton T. Morrison, Mike Cafarella
(Proceedings of the 1st Workshop on AI and Scientific Discovery: Directions and Opportunities (ACL))
- Neural-Guided Program Synthesis of Information Extraction Rules Using Self-Supervision** Jan 2022
Enrique Noriega-Atala, Robert Vacareanu, Gus Hahn-Powell, Marco A. Valenzuela-Escárcega
(Proceedings of the Workshop on Pattern-based Approaches to NLP in the Age of Deep Learning at COLING)
- Learning Open Domain Multi-hop Search Using Reinforcement Learning** Jan 2022
Enrique Noriega-Atala, Mihai Surdeanu, Clayton T. Morrison
(Proceedings of the Workshop on Structured and Unstructured Knowledge Integration (ACL))
- Low Resource Causal Event Detection from Biomedical Literature** Jan 2022
Zhengzhong Liang, *Enrique Noriega-Atala*, Clayton Morrison, Mihai Surdeanu
(Proceedings of the 21st Workshop on Biomedical Language Processing at ACL)
- Neural Architectures for Biological Inter-Sentence Relation Extraction** Jan 2022
Enrique Noriega-Atala, Peter M. Lovett, Clayton T. Morrison, Mihai Surdeanu
(Proceedings of Scientific Document Understanding at AAAI)

Understanding the Polarity of Events in the Biomedical Literature: Deep Learning vs. Linguistically-informed Methods <i>Enrique Noriega-Atala, Zhengzhong Liang, John A. Bachman, Clayton T. Morrison, Mihai Surdeanu</i> (Proceedings of the Workshop on Extracting Structured Knowledge from Scientific Publications at NAACL-HLT)	Jan 2019
Inter-sentence Relation Extraction for Associating Biological Context with Events in Biomedical Texts <i>Enrique Noriega-Atala, Paul D. Hein, Shraddha S. Thumsi, Zechy Wong, Xia Wang, Clayton T. Morrison</i> (Proceedings of the Sixth Workshop on Data Mining in Biomedical Informatics and Healthcare at ICDM)	Jan 2018
Focused Reading: Reinforcement Learning for What Documents to Read <i>Enrique Noriega-Atala, Marco A. Valenzuela-Escárcega, Clayton T. Morrison, Mihai Surdeanu</i> (Proceedings of the Interactive Machine Learning and Semantic Information Retrieval Workshop at ICML)	Jan 2017
Large-scale Automated Reading with Reach Discovers New Cancer Driving Mechanisms <i>Marco A. Valenzuela-Escárcega, Özgün Babur, Gus Hahn-Powell, Dane Bell, Thomas Hicks, Enrique Noriega-Atala, Xia Wang, Mihai Surdeanu, Emek Demir, Clayton T. Morrison</i> (Proceedings of the BioCreative VI Workshop (BioCreative6 2017))	Jan 2017

Conference Presentations

When and Where Did it Happen? An Encoder-Decoder Model to Identify Scenario Context Workshop on the Future of Event Detection at EMNLP 2024	Miami, Florida Jan 2024
Visualizing Interaction Networks and Evidence in Biomedical Corpora The 16th IEEE Pacific Visualization Symposium (PacificVis)	Seoul, South Korea Jan 2023
Neural-Guided Program Synthesis of Information Extraction Rules Using Self-Supervision Workshop on Pattern-based Approaches to NLP in the Age of Deep Learning at COLING	Gyeongju, Republic of Korea Jan 2022
Open Domain Multi-hop Search Using Reinforcement Learning Workshop on Structured and Unstructured Knowledge Integration at NAACL-HLT	Seattle, Washington Jan 2022
Low Resource Causal Event Detection from Biomedical Literature 21st Workshop on Biomedical Language Processing at ACL	Dublin, Ireland Jan 2022
Neural Architectures for Biological Inter-Sentence Relation Extraction Scientific Document Understanding at AAAI	Vancouver, Canada Jan 2022
Understanding the Polarity of Events in the Biomedical Literature: Deep Learning vs. Linguistically-informed Methods Workshop on Extracting Structured Knowledge from Scientific Publications at NAACL-HLT	Minneapolis, Minnesota Jan 2019
Inter-sentence Relation Extraction for Associating Biological Context with Events in Biomedical Text Sixth Workshop on Data Mining in Biomedical Informatics and Healthcare at ICDM	Singapore Jan 2018
Grounding Gradable Adjectives through Crowd-sourcing Language Resources and Evaluation Conference	Miyazaki, Japan Jan 2018

Learning what to read: Focused machine reading

NLP Applications track, EMNLP

Copenhagen, Denmark

Jan 2017

Other Presentations

AZX Chat Bot: An LLM-driven Application Implementation to Support Resilience Against Climate Change

Resilience Informatics Workshop

The University of Arizona
Nov 2023

Visualizing Interaction Networks and Evidence in Biomedical Corpora

Immune Monitoring Symposium

The University of Arizona
Nov 2023

Machine Learning and NLP for Medical Research and Public Health

PHIRE Side Chat

The University of Arizona
May 2023

Accelerating Research through Large Scale, Interpretable Machine Reading

MCB Joint Seminar

The University of Arizona
Oct 2022

Accelerating Research through Large Scale, Interpretable Machine Reading

Applied Data Science Speak & Greet Speaker Series

The University of Arizona
Sept 2022

What is Machine Learning?

Guest lecture for Knowledge-based Systems in Organizations, Systems and Industrial Engineering Program at Monterrey Institute of Technology and Higher Education

Hermosillo, Mexico
Jan 2015

Service

Organization Committee Member

- The Second Workshop in Pattern-based Approaches to NLP in the Age of Deep Learning (Pan-DL), EMNLP (2023)
- Pattern-based Approaches to NLP in the Age of Deep Learning (Pan-DL), COLING (2022)

Program Committee Member

- The AAAI-22 Workshop on Scientific Document Understanding, AAAI (2022)
- SUKI: Structured and Unstructured Knowledge Integration, NAACL-HLT (2022)
- TextGraphs: Graph-based Algorithms for Natural Language Processing (14, 15, and 16)

Student Volunteer

NAACL-HLT

Minneapolis, Minnesota
Jan 2019

Grants

SKEMA: Scientific Knowledge Extraction and Model Analysis

Co-PI, DARPA (Award amount: \$9,345,747)

July 2022 – Dec 2025

Resilience Informatics for Public Health

Key Personnel, Technology and Research Initiative Fund (TRIF), The University of Arizona (Award amount: \$95,000)

July 2023 – June 2024

Patents

Method for Extracting and Assessing Signaling Pathways from Biomedical Literature

Provisional US patent filed. Authors: Mihai Surdeanu, Marco A. Valenzuela-Escarcega, Gustavo Hahn-Powell, Dane Bell, Thomas Hicks, Enrique Noriega, Clayton Morrison

Mar 2017